

TALHA ISLAM

Full Stack AI Engineer

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ABOUT ME

Full Stack AI Engineer with over 4 years building production systems for startups and enterprise teams, currently at Wortholic and earlier with SimpleTalk AI, Inexor, Voliom and SuperAnnotate. I work across the whole path rather than one slice of it: the model and retrieval layer, the Python services behind it, the API contract, the interface that consumes it and the infrastructure it runs on. I have led a group of 7 to 8 engineers, carried the delivery dates myself, and I go after the harder problems on purpose because that is the work I enjoy. What I am building towards is principal level ownership, where the architecture calls and the quality of what ships are mine to answer for.

HOW I WORK

- I read the documentation before I write anything. Provider limits, library internals and failure modes get checked against the source instead of guessed at, so I can defend a design decision when someone questions it.
 - AI is a tool I drive, not a shortcut I hide behind. I run several Claude sessions in parallel, one per workstream, keep each scoped to a single problem, and read every line before it lands in a branch.
 - Frontend, backend, API design, third party integration, data layer or deployment, I pick up whichever part is blocking the release and close it out.
 - Planning, tickets, estimates and status sit in ClickUp so the team can see where a piece of work stands without having to ask anyone.
 - Client systems, credentials and customer data stay under least privilege. Secrets never sit in a repository and access is scoped per environment.
 - Dates are commitments. If scope has to move I say so early rather than let a deadline slip quietly.
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WORK EXPERIENCE

Sr. AI Engineer | [Wortholic](#)

Aug 2025 – Present

AI Engineering Lead / AI Solutions Architect

- Own the architecture for every AI product the company ships, from first sketch through release and the support that follows it.
- Built voice agents, chat assistants, retrieval systems and long running automations that carry live customer traffic.
- Wrote the service layer in FastAPI using async routes, Pydantic schemas, dependency injection, background workers and streamed responses for calls in progress.
- Shipped call centre platforms covering telephony, routing, transcription, escalation and reporting.
- Lead 7 to 8 people across engineering and data science, run the planning board, and review what goes to production.

Tech Stack: FastAPI, LangChain, GPT-4, Claude, Pinecone, Twilio, PostgreSQL, Redis, Docker, Kubernetes, Terraform, React.

Senior AI Engineer | [Simpletalk.ai](#)

Jan 2025 – Jul 2025

Built automation products that let business teams hand repetitive process work over to language model agents.

- **Agent runtime:** agents that plan and execute jobs across several steps, with retries, timeouts and a human approval gate on anything sensitive.
- **Connectors:** integration layer between the runtime and the software clients already ran, covering auth flows, pagination, webhooks and rate limits.
- **Product surface:** interfaces that let a non technical user configure a workflow and watch it run without reading a log file.

Tech Stack: Python, Celery, n8n, webhook orchestration, OpenAI API, function calling, Next.js.

Data Scientist | [Inexor.ai](#)

Mar 2024 – Dec 2024

Built the retrieval and market data side of WealthNx.AI, where an answer that is stale or wrong is worse than no answer.

- **Retrieval:** chunking, embedding and reranking tuned for filings and financial news, with a source attached to every claim the assistant makes.
- **Market models:** time series work on crypto and equities, plus ingestion that keeps prices and fundamentals current across several horizons.
- **Reporting:** dashboards where a user can move from a headline figure down to the transactions underneath it.

Tech Stack: MCP, FAISS, pandas, NumPy, Kafka, distributed processing, Plotly, D3.js.

Data Analyst | [Voliom](#)

Jan 2022 – Jan 2024

Turned scattered company data into something a business team could act on, using early agent tooling and knowledge graphs.

- Designed pipelines and semantic search that covered both structured tables and loose documents.
- Shipped forecasting, anomaly detection and segmentation models that fed the daily reporting cycle.
- Handled voice and text processing for operational reporting close to real time.

Tech Stack: SQL, Snowflake, Airflow, Spark, ETL, knowledge graphs, scikit-learn, Power BI, Tableau.

ADDITIONAL EXPERIENCE

Data Annotator | [SuperAnnotate](#) (Part-time concurrent)

Jan 2024 – Jun 2025

Response evaluation for language model quality assurance, alongside mentoring the interns on the same projects.

- Reviewed over 50,000 model responses while holding 98% agreement with the gold standard.
- Wrote the labelling guidelines that three client projects went on to adopt.
- Cut error rates by 45% by retraining the team on the edge cases they kept missing.

EDUCATION

Bachelor's Degree in Computer Science | [Riphah International University](#)

Jan 2019 – Feb 2023

Spent the degree shipping software rather than only sitting exams. Delivered a financial assistant and Virtual Psych, a mental health platform built for scale, both with real backend services and integrations behind them. Focus on architecture, deployment and owning a product from idea to release. Microsoft Certified Professional in enterprise application development.

Activities: Programming Club, AI & Robotics Society, Hackathons, Tech Meetups.

Key Achievements

- **NaturalFlow.AI** is a business communication platform running inbound and outbound voice agents, SMS agents and website, WhatsApp and Messenger chatbots as a complete call centre. I architected the speech to text, orchestration, model and text to speech path, along with call routing, lead qualification, appointment booking and the CRM hookups into Salesforce, HubSpot, Twilio, Google Calendar, Zapier and Make. Backend holds 100K+ concurrent agents across multiple languages with live call analytics.
- **WealthNx.AI** is a personal finance app on iOS and Android that pulls bank, brokerage and crypto accounts through Plaid across 12,000+ institutions. I engineered the market data pipelines feeding 70,000+ stocks and ETFs and 5 million+ crypto assets from Nasdaq, Benzinga, FMP and CoinGecko, exposed 50+ low latency endpoints and brought infrastructure spend down 35%. Also worked on Wealth Genie, the assistant set that answers accounting, stock and crypto questions and builds custom reports on top of read only, encrypted account access held to ISO 27001.
- **Galileo AI** is the evaluation and observability platform AI teams use to measure what their agents actually do. I contributed to eval pipelines for retrieval and agent behaviour, tracing that pinpoints the exact step where a run went wrong, and the route from offline evaluators to production guardrails so a score can gate a tool call at runtime instead of surfacing in a report a day later.
- **Robin AI** is a legal intelligence platform for enterprise contract teams. I worked on question answering across large contract sets, search that recognises contract type, signatures and key terms without being told where to look, and obligation tracking that converts renewal windows, payment dates and reporting duties into checklists with alerts attached.

- **Genie AI** is a drafting and negotiation product for commercial teams handling their own agreements. I worked on playbook driven review, redlining inside tracked changes, deals that span several linked documents, table extraction across hundreds of files at once, and the connector layer reaching email, CRM, cloud storage and eSign through both API and MCP.
- **Vaikora**, now part of Data443, is a runtime control gateway that sits inline between an agent and the model, tool, database or API it is trying to reach. Every proposed action is scored against a deterministic policy engine and then allowed, constrained, logged or blocked before it executes, and each verdict is signed into an append only audit chain a compliance reviewer can replay. My freelance engineering covered securing that agent traffic and hardening the surrounding system.
- **Nami Navigator** is a location based healthcare discovery system I built on FAISS retrieval with sentiment scoring, deployed on AWS and holding 95%+ recommendation accuracy.
- **Mindy Support** engaged me as a prompt engineer on AI governance for Fortune 500 financial systems, working with the OpenAI API, LangChain and PromptLayer against ISO 42001, NIST and EU AI Act requirements.

TECHNICAL SKILLS

AI & ML: LLMs, prompt engineering, retrieval augmented generation, multi agent systems, tool calling, evaluation and guardrails, fine tuning, NLP, transformers, computer vision, model optimisation

Python: FastAPI (async routes, Pydantic models, dependency injection, background tasks, WebSockets, streaming responses, generated OpenAPI docs), Flask, Django, asyncio, Celery, SQLAlchemy, pytest, type hints, packaging

Backend: Node.js, REST and GraphQL, microservices, JWT and OAuth2, rate limiting, caching, message queues, versioned API design, scalable deployments

ML Tools: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, pandas, NumPy, OpenCV, Hugging Face, LangChain, FAISS, Pinecone

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Terraform, CloudFormation, Prometheus, Grafana

Data & Analytics: SQL, PostgreSQL, MongoDB, Snowflake, Airflow, Kafka, Spark, ETL, data modeling, Plotly, Tableau, Power BI

Automation: n8n, Make, Zapier, GHL, ClickUp, RPA, BPA, webhooks, API orchestration, email and SMS flows

Voice & Chat AI: voice agents, chatbots, IVR, STT/TTS, Twilio, Dialogflow, Lex, Azure Speech

Frontend: React, Next.js, Flutter, Tailwind, Chakra UI, responsive dashboards

Soft Skills: leadership, mentorship, client consulting, collaboration, Agile, Slack, Youtrack

CERTIFICATIONS

- BCG - Data Science Job Simulation | Forage | Apr 2025 | [Link](#)
- Accenture North America - Data Analytics and Visualization | Forage | Aug 2024 | [Link](#)
- Accenture North America - Project Management Job Simulation | [Link](#)
- IBM Artificial Intelligence | [Link](#)
- IBM Supervised Machine Learning: Classification | [Link](#)

ADDITIONAL INFORMATION

Languages: English, Urdu

Interests: AI Innovation, Gaming